MOHAMED NADIM TOUIL

Robotics Software Engineer

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PROFILE

Robotics software engineer building ROS 2 autonomy and simulation for legged and other robotic platforms, with autonomy code in production on a commercial quadruped fleet and hands-on hardware deployment on marine and ground robots. RL locomotion and vision-guided manipulation in simulation.

EDUCATION

National Engineering Degree — Advanced Technologies 2021 – 2024
ENSTAB, University of Carthage, Tunisia *Specialization: Embedded Systems, Control & Robotics*

Preparatory Institute for Engineering Studies (IPEIM), University of Monastir, Tunisia 2019 – 2021

EXPERIENCE

Robotics & AI Software Engineer, EagleProjects Tunisia — Tunis, Tunisia May 2025 – Present
ROS 2 autonomy stack for a client fleet of about 45 Unitree quadrupeds; hardware deployment by the Italian team.
- Built frontier-based autonomous exploration as a ROS 2 service: 8-neighbour frontier detection, KD-tree clustering, distance-vs-size medoid scoring; a 7-min run covered 58 m across 8 frontier goals in sim
- Built LiDAR reactive obstacle avoidance as ROS 2 actions: rotate-then-wall-follow and lateral-slide strategies, and live obstacle injection into the A* planner for local replanning
- Developed the Gazebo simulator for Go2, A2, A2W, B2 and B2W with a bridge layer mirroring the real Unitree topic interface, so the production stack runs unchanged; used by clients to validate missions before hardware
- Trained G1 motion-imitation policies from video with the open-source PBHC pipeline (IsaacGym, PPO; MuJoCo sim2sim); hardware deployment handled by the partner team

ROS & Robotics Mentor (Part-time), Graine d'Entrepreneur — Tunis, Tunisia Feb – Jun 2025
Coached interns through end-to-end ROS 1/2 autonomous systems: sensor integration, URDF modeling, and navigation stack configuration.

Robotics Engineering Intern, VGR Laboratory, York University — Toronto, Canada May – Aug 2024
- Built and deployed an autonomous surface vessel in ROS 2 on physical hardware, integrating LiDAR and camera for open-water navigation
- Implemented ArUco-based visual docking with clothoid trajectory planning: repeatable docking at 20% and 50% thrust, failure envelope characterized at 80%

PROJECTS

hand-teleop — ROS 2 Humble, MuJoCo, MediaPipe, C++ differential IK 2026
Webcam hand teleoperation of a UR5e + Robotiq 2F-85 with a 200 Hz differential IK controller: 20/20 pick-and-place, 62 ms median latency, 5.1 mm RMS tracking.

tog-sim — ROS 2 Humble, Gazebo Fortress, Ruckig, YOLO11-seg 2026
Vision-guided high-speed SCARA pick-and-place cell: segmentation model trained on 1,773 synthetic scenes (mask mAP50 0.98); picks from moving belts at 12–14 picks/min, 2.7 mm mean placement, 80/80 success.

CartPole-Goes-Brrr — PyBullet, Gymnasium, Stable-Baselines3 2026
Four-task PPO curriculum on one physics rig using reward shaping only: cartpole and double-pendulum balance, inchworm crawl (5 m/ep), gantry hopping (16.9 m/ep).

Competitive Autonomous Robots — National competitions 2022 – Present
PID line-followers with sensor fusion and real-time decision logic; one first and three second places in national competitions.

SKILLS

Robotics & Autonomy: ROS 2 Humble, Nav2 (AMCL, map_server), MoveIt 2, SLAM (slam_toolbox, gmapping), behavior trees, state machines, Unitree SDK2 / DDS, ArUco vision, sensor fusion
Simulation & RL: Gazebo (Classic), MuJoCo, PyBullet, RViz, Gymnasium, Stable-Baselines3, PPO, reward shaping, sim curricula
Perception: LiDAR (RoboSense Helios), Intel RealSense D435i, OpenCV, YOLO (Ultralytics), MediaPipe
Programming: Python (primary); C++, C (academic projects)
Embedded & Tools: ESP32, STM32, Raspberry Pi, Jetson, Arduino; Linux, Git, Docker, Jira
Languages: Arabic (Native), English (Fluent), French (Fluent)